

Engineering Economics Questions

Q1: Given a Project with the following data: First (initial) cost = JD 7000; Annual Operating cost = JD 1200; Salvage Value = JD 1500; $i = 8\%$ per year, and $n = 12$ years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 4, we use:

• $7000(F/P, 8\%, 3) + 1200(P/A, 8\%, 12)(F/P, 8\%, 3) - 1500(P/F, 8\%, 8)$

• $7000(F/P, 8\%, 4) + 1200(P/A, 8\%, 12)(F/P, 8\%, 4) - 1500(P/F, 8\%, 8)$

• $7000(F/P, 8\%, 4) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 8)$

Correct Answer: c. $7000(F/P, 8\%, 4) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 8)$

Q2: A project with an annual operating cost JD 1000, increased by 100 yearly over $n = 12$ years. If $i = 8\%$ per year, then the equivalent single payment at time zero (Eq. $X(0)$) is around:

• 10165.4

• 3463.3

• 10999.3

Correct Answer: 10999.3

Q3: Given a Salvage Value = JD 1000; $i = 8\%$ per year and $n = 11$ years. Then, the Eq. $X(7)$ (equivalent single value at the end of year 7) is:

• -1360.4

• -735.03

• -428.88

Correct Answer: -428.88

Q4: The missing Factor in the equation $(F/A, i, n) (\text{---?---}) = (P/A, i, n)$ is:

• $(F/A, i, n)$

• $(A/F, i, n)$

• $(P/F, i, n)$

Correct Answer: $(P/F, i, n)$

Q5: An interest rate of 5% per month compounded continuously is the same as:

• an effective 5% per month.

• an effective 2.5% per quarter.

• an effective 5.12% per month.

Correct Answer: an effective 5.12% per month.

Q6: An interest rate of 10% per year compounded semiannually is the same as:

• an effective 5% per quarter compounded continuously.

• an effective 10.25% per year.

• an effective 10% per month.

Correct Answer: an effective 10.25% per year.

Q7: A project with an annual operating cost = JD 1000, increased by 100 yearly over $n = 6$ years. If $i = 8\%$ per year, then the equivalent single payment at time zero (Eq. $X(0)$) is around:

• 4729.94

• 5675.1

• 1052.3

Correct Answer: 4729.94

Q8: Given a Salvage Value = JD 1000; $i = 8\%$ per year and $n = 11$ years. Then, the Eq. $X(8)$ (equivalent single value at the end of year 8) is:

• -428.88

• -793.83

• -1259.7

Correct Answer: -793.83

Q9: A project with an annual operating cost = JD 1000, increased by 100 yearly over $n = 4$ years. If $i = 8\%$ per year, then the equivalent single payment at time zero (Eq. $X(0)$) is around:

• 2821.6

• 244.5

• 3777.1

Correct Answer: 2821.6

Q10: A project with an annual operating cost = JD 1000, increased by 100 yearly over $n = 13$ years. If $i = 8\%$ per year, then the equivalent single payment at time zero (Eq. $X(0)$) is around:

• 3904.6

• 10999.3

• 11808.3

Correct Answer: 11808.3

Q11: For an original (first) amount of JD 1500, at a Simple Interest Rate $i = 10\%$ per year, the Interest Amount (I) after 2 years is:

• JD 300

• JD 1800

• JD 30 000

Correct Answer: JD 300

Q12: Given a Project with the following data: First (initial) cost = JD 7000; Annual Operating cost = JD 1200; Salvage Value = JD 1500; $i = 8\%$ per year, and $n = 12$ years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 6, we use:

• $7000(F/P, 8\%, 6) + 1200(P/A, 8\%, 12)(F/P, 8\%, 6) - 1500(P/F, 8\%, 6)$

• $7000(F/P, 8\%, 6) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 6)$

• $7000(F/P, 8\%, 5) + 1200(P/A, 8\%, 12)(F/P, 8\%, 5) - 1500(P/F, 8\%, 6)$

Correct Answer: $7000(F/P, 8\%, 6) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 6)$

Q13: The missing Factor in the equation $(P/F, i, n) (---?---) = (A/F, i, n)$ is:

• $(A/P, i, n)$

• $(P/A, i, n)$

• $(F/P, i, n)$

Correct Answer: $(A/P, i, n)$

Q14: An interest rate of 12% per semiannually compounded monthly is the same as:

• an effective 1.5% per month.

• an effective 26.8% per year.

• an effective 12.10% per year.

Correct Answer: an effective 12.10% per year.

Q15: The missing factor in the equation $(A/G, i, n) (P/A, i, n) = (---?---)$ is:

• $(G/P, i, n)$

• $(P/G, i, n)$

• $(A/P, i, n)$

Correct Answer: $(P/G, i, n)$

Q16: For an original (first) amount of JD 1500, at a Simple Interest Rate $i = 10\%$ per year, the Interest Amount (I) after 9 years is:

• JD 135 000

• **JD 1350**

• JD 2850

Correct Answer: JD 1350

Q17: For an original (first) amount of JD 1500, at a Simple Interest Rate $i = 10\%$ per year, the Interest Amount (I) after 7 years is:

• **JD 1050**

• JD 2550

• JD 105 000

Correct Answer: JD 1050

Q18: Given a Salvage Value = JD 1000; $i = 8\%$ per year and $n = 11$ years. Then, the Eq. $X(1)$ (equivalent single value at the end of year 1) is:

• -463.2

• **-1080**

• -428.88

Correct Answer: -1080

1. Given a Project with the following data: First (initial) cost = JD 7000; Annual Operating cost = JD 1200; Salvage Value = JD 1500; $i = 8\%$ per year, and $n = 12$ years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 4, we use:

$$\bullet 7000(F/P, 8\%, 3) + 1200(P/A, 8\%, 12)(F/P, 8\%, 3) - 1500(P/F, 8\%, 8)$$

$$\bullet 7000(F/P, 8\%, 4) + 1200(P/A, 8\%, 12)(F/P, 8\%, 4) - 1500(P/F, 8\%, 8)$$

$$\bullet 7000(F/P, 8\%, 4) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 8)$$

2. Given a Project with the following data: First (initial) cost = JD 7000; Annual Operating cost = JD 1200; Salvage Value = JD 1500; $i = 8\%$ per year, and $n = 12$ years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 2, we use:

$$\bullet 7000(F/P, 8\%, 2) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 10)$$

$$\bullet 7000(F/P, 8\%, 2) + 1200(P/A, 8\%, 12)(F/P, 8\%, 2) - 1500(P/F, 8\%, 10)$$

$$\bullet 7000(F/P, 8\%, 1) + 1200(P/A, 8\%, 12)(F/P, 8\%, 4) - 1500(P/F, 8\%, 10)$$

3. Given a Project with the following data: First (initial) cost = JD 7000; Annual Operating cost = JD 1200; Salvage Value = JD 1500; $i = 8\%$ per year, and $n = 12$ years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 5, we use:

$$\bullet 7000(F/P, 8\%, 5) + 1200(P/A, 8\%, 12)(F/P, 8\%, 5) - 1500(P/F, 8\%, 7)$$

$$\bullet 7000(F/P, 8\%, 5) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 7)$$

$$\bullet 7000(F/P, 8\%, 4) + 1200(P/A, 8\%, 12)(F/P, 8\%, 4) - 1500(P/F, 8\%, 7)$$

4. Given a Project with the following data: First (initial) cost = JD 7000; Annual Operating cost = JD 1200; Salvage Value = JD 1500; $i = 8\%$ per year, and $n = 12$ years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 6, we use:

$$\bullet 7000(F/P, 8\%, 6) + 1200(P/A, 8\%, 12)(F/P, 8\%, 6) - 1500(P/F, 8\%, 6)$$

$$\bullet 7000(F/P, 8\%, 6) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 6)$$

$$\bullet 7000(F/P, 8\%, 5) + 1200(P/A, 8\%, 12)(F/P, 8\%, 5) - 1500(P/F, 8\%, 6)$$

5. Given a Project with the following data: First (initial) cost = JD 7000; Annual Operating cost = JD 1200; Salvage Value = JD 1500; $i = 8\%$ per year, and $n = 12$ years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 7, we use:

$$\bullet 7000(F/P, 8\%, 7) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 5)$$

$$\bullet 7000(F/P, 8\%, 7) + 1200(P/A, 8\%, 12)(F/P, 8\%, 7) - 1500(P/F, 8\%, 5)$$

$$\bullet 7000(F/P, 8\%, 6) + 1200(P/A, 8\%, 12)(F/P, 8\%, 6) - 1500(P/F, 8\%, 5)$$

6. Given a Project with the following data: First (initial) cost = JD 7000; Annual Operating cost = JD 1200; Salvage Value = JD 1500; $i = 8\%$ per year, and $n = 12$ years. To Calculate the Equivalent Single Value (Eq. X) at the end of year 9, we use:

• $7000(F/P, 8\%, 9) + 1200(P/A, 8\%, 12)(F/P, 8\%, 9) - 1500(P/F, 8\%, 3)$

• $7000(F/P, 8\%, 8) + 1200(P/A, 8\%, 12)(F/P, 8\%, 8) - 1500(P/F, 8\%, 3)$

• $7000(F/P, 8\%, 9) + 1200(P/A, 8\%, 12) - 1500(P/F, 8\%, 3)$

7. A project with an annual operating cost = JD 1000, increased by 100 yearly over $n = 12$ years. If $i = 8\%$ per year, then the equivalent single payment at time zero (Eq. $X(0)$) is around:

• 10165.4

• 3463.3

• 10999.3

8. A project with an annual operating cost = JD 1000, increased by 100 yearly over $n = 4$ years. If $i = 8\%$ per year, then the equivalent single payment at time zero (Eq. $X(0)$) is around:

• 2821.6

• 244.5

• 3777.1

9. A project with an annual operating cost = JD 1000, increased by 100 yearly over $n = 6$ years. If $i = 8\%$ per year, then the equivalent single payment at time zero (Eq. $X(0)$) is around:

• 4729.94

• 5675.1

• 1052.3

10. A project with an annual operating cost = JD 1000, increased by 100 yearly over $n = 7$ years. If $i = 8\%$ per year, then the equivalent single payment at time zero (Eq. $X(0)$) is around:

• 1402.4

• 5675.1

• 6608.7

11. A project with an annual operating cost = JD 1000, increased by 100 yearly over $n = 8$ years. If $i = 8\%$ per year, then the equivalent single payment at time zero (Eq. $X(0)$) is around:

• 7527.2

• 1780.6

• 6608.7

12. A project with an annual operating cost = JD 1080, increased by 100 yearly over $n = 11$ years. If $i = 8\%$ per year, then the equivalent single payment at time zero (Eq. $X(0)$) is around:

• 10165.4

• 3026.5

• 9307.6

13. A project with an annual operating cost = JD 1000, increased by 100 yearly over $n = 13$ years. If $i = 8\%$ per year, then the equivalent single payment at time zero (Eq. $X(0)$) is around:

• 3904.6

• 10999.3

• 11808.3

14. Given a Salvage Value = JD 1000; $i = 8\%$ per year and $n = 11$ years. Then, the Eq. $X(1)$ (equivalent single value at the end of year 1) is:

• -463.2

• -1080

• -428.88

15. Given a Salvage Value = JD 1000; $i = 8\%$ per year and $n = 11$ years. Then, the Eq. $X(4)$ (equivalent single value at the end of year 4) is:

• -428.88

• -583.49

• -1713.8

16. Given a Salvage Value = JD 1000; $i = 8\%$ per year and $n = 11$ years. Then, the Eq. $X(5)$ (equivalent single value at the end of year 5) is:

• -428.88

• -630.17

• -11586.8

17. Given a Salvage Value = JD 1000; $i = 8\%$ per year and $n = 11$ years. Then, the Eq. $X(7)$ (equivalent single value at the end of year 7) is:

• -428.88

• -735.03

• -1360.4

18. Given a Salvage Value = JD 1000; $i = 8\%$ per year and $n = 11$ years. Then, the Eq. X(8) (equivalent single value at the end of year 8) is:

• -428.88

• -793.83

• -1259.7

19. Given a Salvage Value = JD 1000; $i = 8\%$ per year and $n = 11$ years. Then, the Eq. X(9) (equivalent single value at the end of year 9) is:

• -857.34

• -1166.4

• -428.88

20. Given a Salvage Value = JD 1000; $i = 8\%$ per year and $n = 11$ years. Then, the Eq. X(10) (equivalent single value at the end of year 10) is:

• -1080

• -925.93

• -428.88

21. For an original (first) amount of JD 1500, at a Simple Interest Rate $i = 10\%$ per year, the Interest Amount (I) after 2 years is:

• JD 300

• JD 1800

• JD 30 000

22. For an original (first) amount of JD 1500, at a Simple Interest Rate $i = 10\%$ per year, the Interest Amount (I) after 6 years is:

• JD 90 000

• JD 900

• JD 2400

23. For an original (first) amount of JD 1500, at a Simple Interest Rate $i = 10\%$ per year, the Interest Amount (I) after 7 years is:

• JD 1050

• JD 2550

• JD 105 000

24. For an original (first) amount of JD 1500, at a Simple Interest Rate $i = 10\%$ per year, the Interest Amount (I) after 9 years is:

• JD 2850

• **JD 1350**

• JD 135 000

25. For an original (first) amount of JD 1500, at a Simple Interest Rate $i = 10\%$ per year, the Interest Amount (I) after 12 years is:

• JD 3300

• JD 180 000

• **JD 1800**

26. The missing Factor in the equation $(F/A, i, n) (\text{---?---}) = (P/A, i, n)$ is:

• $(F/A, i, n)$

• $(A/F, i, n)$

• **$(P/F, i, n)$**

27. The missing Factor in the equation $(P/F, i, n) (\text{---?---}) = (A/F, i, n)$ is:

• $(P/A, i, n)$

• $(F/P, i, n)$

• **$(A/P, i, n)$**

28. The missing factor in the equation $(A/G, i, n) (P/A, i, n) = (\text{---?---})$ is:

• **$(P/G, i, n)$**

• $(G/P, i, n)$

• $(A/P, i, n)$

29. The missing Factor in the equation $(\text{---?---}) (F/P, i, n) = (F/A, i, n)$ is:

• $(P/A, i, n)$

• $(F/P, i, n)$

• **$(F/A, i, n)$**

30. The missing Factor in the equation $(P/F, i, n) (A/P, i, n) = (---?---)$ is:

• **$(A/F, i, n)$**

• $(A/P, i, n)$

• $(P/A, i, n)$

31. An interest rate of 5% per month compounded continuously is the same as:

• an effective 5% per month.

• an effective 2.5% per quarter.

• **an effective 5.12% per month.**

32. An interest rate of 10% per year compounded semiannually is the same as:

• an effective 10% per month.

• **an effective 10.25% per year.**

• an effective 5% per quarter compounded continuously.

33. An interest rate of 8% per year compounded monthly is the same as:

• **an effective 8.29% per year.**

• an effective 8% per year.

• an effective 8.29% per month.

34. What is the conductance of an ammeter that indicates 20 A when 0.01 V is across it?

• 0 S

• **2000 S**

• 20 S

• 200 S

35. If a 12V car battery has a 0.1 ohm internal resistance, what terminal voltage causes a 4A current to flow into the positive terminal?

• 16 V

• **12.4 V**

• 12 V